THOMAS LABETOUILLE

Simulation, real-time software and validation

Toulouse, France • +33 6 52 62 96 37 • thomas.labetouille@gmail.com • Driving licence, own car

LinkedIn • Portfolio • GitHub

Kalman filters	PX4 drones	C++ · Python · Rust	14 months
sensor fusion	real flight logs	embedded, automated tests	aeronautical simulation — defence

PROFILE

I build software that lets a drone know where it is from its sensors (GPS, inertial unit, magnetometer, barometer, airspeed), and that keeps working when one of them fails. Before that, 14 months at CS Group on aeronautical simulators for defence. Available immediately.

DRONE PROJECTS — *public code on GitHub*

Drone navigation by sensor fusion — Python, C++ 2026

github.com/ThomasLabetouille/drone-nav-estimation

- **Extended Kalman filter (EKF)** combining 5 sensors to estimate the drone's position, velocity and orientation.
- **Robust to failures**: after 60 s without GPS, 15 m of error, against 85 m with the inertial unit alone.
- **Tested on 3 real PX4 drone flights**: less than half a metre from the onboard autopilot.
- **Rewritten in embedded C++**, with results identical to the Python version; 197 automated tests.

Vigie — drone flight safety — Rust 2026

github.com/ThomasLabetouille/vigie

- **Embedded software** that monitors a drone (authorised area, battery, ground link) and brings it back on its own when something goes wrong.
- Tested in **PX4 / Gazebo simulation**; core compatible with microcontrollers (Cortex-M).

WORK EXPERIENCE

Embedded Systems Technician — CS Group Sep. 2024 – Nov. 2025 · Toulouse

Inscape VTS aeronautical simulators — national defence programme

- Designed **real-time training scenarios** for flight simulators, under strict requirements.
- Worked directly with systems engineers and pilots, in a regulated defence environment.

Gameplay Programmer C++ (Freelance) — Rainbow Ant Studio Dec. 2023 – Sep. 2024

Video game on Unreal Engine 5, remote team — character mechanics in C++

OTHER PROJECTS

Automated testing tools and extensions for 3D engines (Unity, Unreal Engine 5), rule-constrained AI assistant. Details on the portfolio.

SKILLS

Navigation	Kalman filters (EKF), sensor fusion, inertial navigation
Drones	PX4, flight log analysis, Gazebo simulation, MAVLink
Languages	C++, Python, Rust, C#
Quality	automated testing, continuous integration (GitHub Actions)

EDUCATION

Bachelor in Video Game Development — E-Artsup 2020 – 2023

RNCP level 6 qualification (French 3-year degree, Bac+3) — programming, software design, real-time engines

ADDITIONAL INFORMATION

French (native) · fluent technical English · French national, eligible for security clearance · boxing and dance